

**Fig. 7.20 *Cannabis sativa* (Hemp plant)**

**Cocaine** is a white powder that is obtained from the leaves of the coca plant, *Erythroxylum coca*. It is commonly called coke or crack. Cocaine causes serious physical and psychological problems including **hallucinations** and **paranoia**. The other plants with hallucinogenic properties are *Atropa belladonna* and *Datura* **(Fig. 7.21 and Fig. 7.22).**



**Fig. 7.21 *Atropa belladonna***



**Fig. 7.22 *Datura***

Drugs like **methamphetamine, amphetamines, barbiturates, tranquilizers, Lysergic acid diethylamide (LSD)** are normally used as medicine to treat patients with mental illness like **depression** and **insomnia** and are often abused.

Tobacco is smoked, chewed and used as snuff. It increases the carbon monoxide content of blood and reduces the concentration of haem bound oxygen, thus causing oxygen deficiency in the body. Tobacco contains **nicotine, carbon monoxide** and **tars,** which cause problems in the heart, lung and nervous system. Adrenal glands are stimulated by nicotine to release adrenaline and nor adrenaline which increases blood pressure and heart beat.

### 7.6.1 Addiction and dependence

Addiction is a physical or psychological need to do or take or use certain substance such as alcohol, to the point where it could be harmful to the individual. This addictive behaviour can be personally destructive to a person. Overtime addicts start to lose not only their jobs, homes and money, but also friendship, family relationships and contact with the normal world. Addiction to drugs and alcohol can lead to a psychological attachment to certain effects such as **euphoria** and temporary feeling of well being.

Repeated use of drugs and alcohol may affect the tolerance level of the receptors present in the body. These receptors then respond only to highest doses of drugs and alcohol leading to greater intake and addiction. Excessive use of drug and alcohol leads to physical and psychological dependence. When psychological dependence develops, the drug user gets mentally **'hooked on'** to the drug. The drug user constantly thinks only about the drug and has continuous uncontrollable craving for it. This state called "**euphoria**" is characterized by mental and emotional preoccupation with the drug.

Physical dependence is a state in which the user's body requires a continuous presence of the drug. If the intake of the drug or alcohol is abruptly stopped, he or she would develop **withdrawal symptoms**. In a sense, the body becomes confused and protests against the absence of the drug. The withdrawal symptoms may range from mild tremors to convulsions, severe agitation and fits, depressed mood, anxiety, nervousness, restlessness, irritability, insomnia, dryness of throat, etc, depending on the type of drug abuse.



XII Std Biology-Zoology Chapter-7 EM.indd 131 23/12/2022 12:40:00